MODEL comparision matrix Test

Classfication	Accuracy	Precision	Recall	F1Score	ROC/AUC
Logistic Regression	0.7635	0.5416	0.7189	0.6177	0.8375
Decision Tree Classifier	0.7531	0.5259	0.7224	0.6087	0.8166
Random Forest Classifier	0.7521	0.5236	0.7509	0.6170	0.8404
Gradient Boosting Classifier (e.g., XGBoost or LightGBM)	0.7711	0.56	0.6477	0.6007	0.8234
K-Nearest Neighbors Classifier	0.7124	0.4724	0.7011	0.5645	0.7790
Support Vector Classifier (SVC)	0.7720	0.5541	0.7295	0.6298	0.8274
Voting Vs Avearge (best 3 out of 6)	0.7720	0.5543	0.7260	0.6287	0.8428
Bayesian Ensemble	0.7720	0.5543	0.7260	0.6287	0.8428

Both voting and Bayesian are close because they both weigh each section at 33%

Which model gave you the best result and which was the worst?

The model that did the best was SVC. However, in the end I chose Random Forest as it has a higher recall.

The worst was KNN

Valadation

Classfication	Accuracy	Precision	Recall	F1Score	ROC/AUC
Logistic Regression	0.75	0.519	0.7786	0.6229	0.8320
Decision Tree Classifier	0.7547	0.5259	0.7607	0.6219	0.8275
Random Forest Classifier	0.7509	0.5192	0.8214	0.6362	0.8447
Gradient Boosting Classifier (e.g., XGBoost or LightGBM)	0.7585	0.5348	0.6857	0.6009	0.8140
K-Nearest Neighbors Classifier	0.7273	0.4910	0.7786	0.6022	0.7993
Support Vector Classifier (SVC)	0.7642	0.5390	0.7643	0.6322	0.8256
Voting Vs Avearge (best 3 out of 6)	0.7595	0.5317	0.7786	0.6319	0.8398
Bayesian Ensemble	0.7585	0.5304	0.7786	0.6310	0.8398